# Scientific Intelligence & Target Discovery

## A Governed Agentic AI Reference Architecture for Life Sciences

*Agent 10 of 8 · synthesize literature, omics & patents into ranked, evidence-linked targets — a scientist owns the hypothesis [roadmap]*

**HCLS AI Agent Suite**

HCLS AI Agent Suite · Built on AWS · June 2026

# FROM A MILLION PAPERS A YEAR TO RANKED TARGETS

## Scientific Intelligence & Target Discovery

*A governed discovery copilot that synthesizes literature, omics, patents, and internal data into ranked, evidence-linked targets with full provenance — while a scientist owns the target hypothesis and every go/no-go decision.*

**~86%**

of drugs entering clinical testing fail; wrong target / lack of efficacy leads

*[gov/peer-reviewed — Wong 2019]*

**11%**

of landmark preclinical cancer studies were reproducible (Amgen)

*[gov/peer-reviewed — Begley 2012]*

**>1M / yr**

new papers added to PubMed (>35M total) — no team can read it all

*[gov/peer-reviewed — NLM]*

# The Issue & The Cost of Doing Nothing

## THE ISSUE

- Target evidence is spread across >35M papers, omics, patents, and internal data — no team can synthesize it manually.
- Poor target validation is a leading cause of the ~86% clinical failure rate.
- The preclinical evidence base is shaky — only ~11% of landmark cancer studies reproduced.
- Discovery-to-candidate runs ~3–6 years and hundreds of millions before a single patient.

## THE COST OF DOING NOTHING

**~$2.6B / drug**

*Industry: ~$2.6B capitalized cost per approved drug (incl. failures); a typical preclinical program is ~$430M out-of-pocket over 3–6 years. Failed targets are the dominant driver.*

- Years and hundreds of millions sunk into targets that fail in the clinic.
- Missed prior art / evidence that a competitor already acted on.

*[industry-research — DiMasi/Tufts; PhRMA]*

**The design line:** the agent synthesizes and ranks evidence with full provenance; a scientist owns the target hypothesis and the go/no-go decision.

# How We Solve It — A Governed Pipeline

*Retrieve literature, omics, patents and internal data through a scoped gateway, synthesize and rank targets with provenance — then route the hypothesis to a scientist.*

**1**
**Retrieve evidence**
literature/omics/patents/internal via gateway + KB

→

**2**
**Synthesize & link**
extract claims, link to source, de-duplicate

→

**3**
**Rank targets**
score on evidence strength + tractability

→

**4**
**HUMAN GATE**
scientist owns the hypothesis & go/no-go

→

**5**
**Append-only audit**
full provenance + decisions logged

### Every action audited

node, timestamp, actor, data sources, model + prompt version — append-only and inspection-ready (21 CFR Part 11).

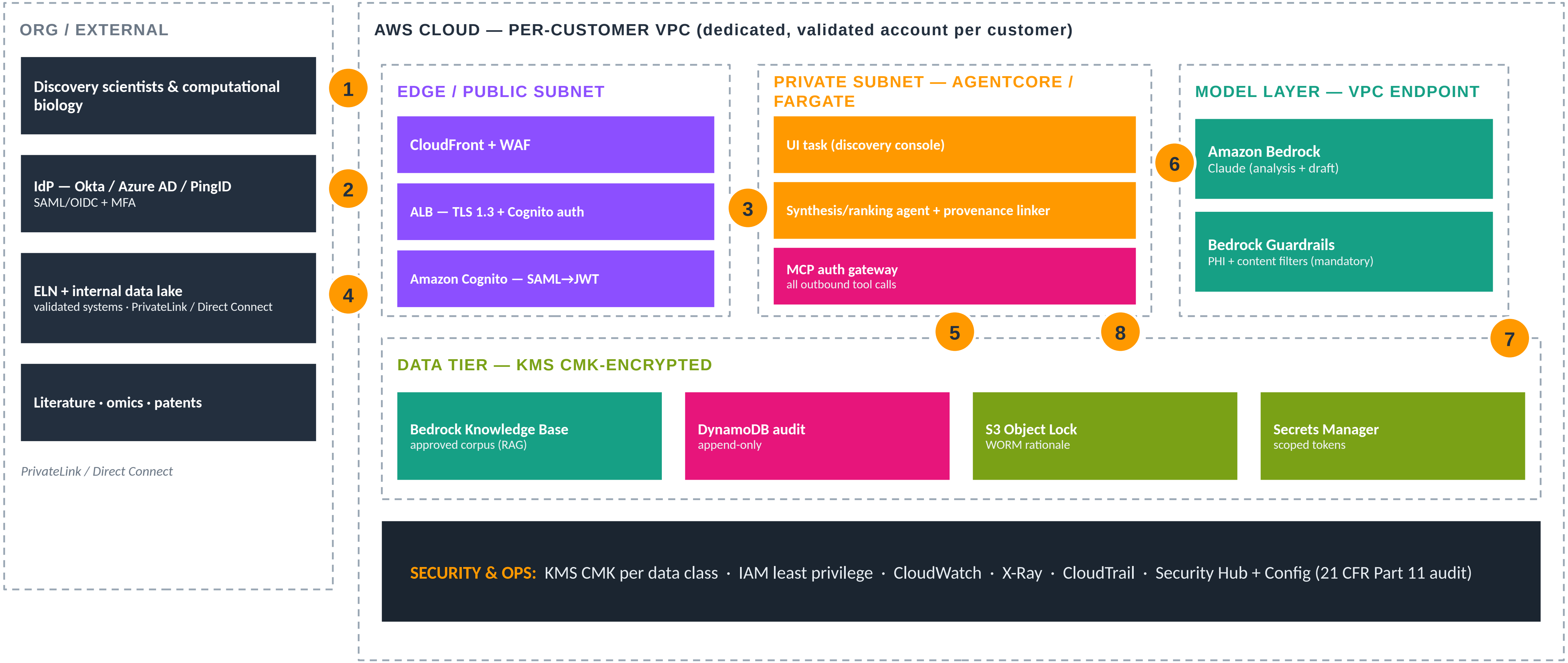
### Grounded & explainable

every regulated claim traces to the source corpus; Bedrock Guardrails mask PHI before any model call and bound the output.

### AI never decides the target hypothesis

the consequential action is a human-only step, enforced in the orchestration framework — not a policy SOP.

# AWS Architecture & Traffic Flow




1 scientist sign-in 2 SAML federation (MFA at IdP; Cognito issues JWT) 3 authenticated session to discovery console 4 ELN + internal data reads over private connectivity (least-privilege) 5 demand scales the agent workers 6 model calls stay in-VPC via Bedrock + Guardrails; internal evidence never egresses after masking 7 full provenance + decisions persisted to append-only audit 8 connectors reachable only through the governed MCP gateway

# Proof, Payback & How to Deploy

## MEASURED OUTCOMES

**~18 mo**

AI novel-target → preclinical candidate (Insilico, company-reported)

*[vendor — company-reported]*

**~50%**

tissue-analysis time cut on AWS (AstraZeneca)

*[vendor — AWS]*

**provenance**

every claim linked to source — reproducibility by design

*[design property]*

**~$430M**

typical preclinical spend the workflow de-risks

*[industry-research]*

## WHAT IT TAKES TO DEPLOY

1. Provision KMS CMK + per-customer validated VPC / network.
1. Stand up Cognito SAML→JWT federation (scientist / comp-bio roles).
1. Front with CloudFront + WAF, ALB TLS 1.3.
1. Deploy the agent + grant tools via the MCP gateway (deny-by-default).
1. Connect ELN + internal data; index literature/omics/patents into a Knowledge Base.
1. Enable S3 Object Lock + append-only audit; run smoke + provenance/gate test.

**Deploy:** *Roadmap agent (Documented): build per docs/CREATE-A-NEW-AGENT.md, then CloudFormation quick-deploy provisions the isolated environment + dual MCP gateway; connect ELN + index the corpus.*

*docs/specs/10-scientific-intelligence.md (design spec — roadmap agent, Documented maturity)*

**The takeaway:** the agent isn't the product — the governed platform that makes it 21 CFR Part 11 / GxP-defensible and deployable on AWS is.